# JULIAN ESTEBAN RINCÓN RODRÍGUEZ

Computer Science and Artificial Intelligence | Data Engineering · Machine Learning · Backend
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## PROFESSIONAL SUMMARY

Final-year (9th semester) Computer Science and Artificial Intelligence student focused on data engineering, machine learning, and backend development. Builds end-to-end data pipelines on AWS, rigorously evaluated machine learning models, and AI agent tooling on top of LLM APIs, with experience delivering software to paying clients and working in teams of three to five. Seeking a six-month data or software engineering internship for January – June 2027.

## EDUCATION

**Universidad Sergio Arboleda** — Bogotá, Colombia | 2022 – Present
B.Sc. Computer Science and Artificial Intelligence | 9th of 9 semesters | GPA 4.04/5.0, with the last three semesters above 4.5/5.0.

Relevant coursework: Big Data and Data Engineering, Machine Learning, Neural Networks, Natural Language Processing, Modeling and Simulation.

## PROFESSIONAL EXPERIENCE

**Full-Stack and AI Developer (Freelance) — Zafra Asesores Tributarios** | Girardot, Colombia | Jun. 2026

- Built a complete sales CRM for a tax advisory firm: FastAPI/SQLAlchemy backend with a full relational model of clients, prospects, campaigns, and interactions, plus a React/Vite frontend for prospect pipeline management.
- Integrated an AI layer using Groq (Llama 3.3 70B) for interaction summaries, email drafting, and lead scoring, along with a qualification bot on the WhatsApp Cloud API.
- Delivered and demoed the system to the client with technical documentation and a defined remaining scope (authentication and continuous deployment) ahead of production rollout.

**Full-Stack Developer — ReushiGo** | Bogotá, Colombia | May 2026
*Full-stack ordering platform for a family coffee distribution business: React/Vite admin panel, Express/Prisma backend, Telegram sales bot · reushi-go.vercel.app*

- Developed the admin panel (React/Vite/Tailwind) and backend (Express/Prisma), with admin-key authentication and rate limiting.
- Implemented a Telegram sales bot with interactive menus and natural-language ordering over a live 19-product catalog.
- Set up CI on GitHub Actions (lint, build, npm audit, tests) and documented a security review of the project.

## SELECTED PROJECTS

**SAVI — Autonomous Real Estate Valuation Agent** | 2026 | v1–v2 team of 4, v3 individual
*Python · XGBoost · PyTorch (Double DQN) · AWS Lambda · EC2 · S3 · SageMaker · RAG · Claude API*

- Audited the academic team version (v2), found that 93% of its dataset was assessor records padded with constant columns, and rebuilt the data from real public sources: 2,930 sales joined to 18,078 parcels, with FHFA and Zillow indices.
- Re-engineered the system individually as an event-driven AWS pipeline (S3 → Lambda → EC2 → Lambda → SageMaker Spot) with boto3 infrastructure as code, automatic shutdown, and a cost under US$0.05 per run; 191 tests and CI on GitHub Actions.
- Reached out-of-fold $R^2$(log) = 0.921 and 7.8% MAPE with XGBoost, and trained RL policies (Value Iteration, Q-Learning, Double DQN) that approve, review, or reject valuations, evaluated once on test with 95% bootstrap intervals.
- Deployed a RAG agent with 8 tools (Claude tool-use loop with a deterministic fallback) and a public web interface on a Lambda Function URL.

**Phantom — Local Multi-Agent Coding Workspace** | Sep. 2026 – Present
*Rust · TypeScript · Next.js · Tauri · MCP · ACP · Whisper · systemd*

- Built, on a fork of the open-source Codeg project (Apache-2.0), a single interface for Claude Code, OpenCode, and Hermes Agent with a unified timeline and delegation between agents via mentions.
- Implemented a model scorecard based on measured session history (95% Wilson interval) that drives delegation and quota failover, and a fully local voice layer with a distinct voice per agent (Whisper large-v3-turbo on CUDA, Chatterbox, and Kokoro).
- Implemented a delegation router that assigns agent, model, and effort level by task type and difficulty, learns from ratings, and switches to a cheaper profile near the quota limit; deployed everything as loopback-only systemd services reachable from a phone only over Tailscale (Telegram with permission buttons), with a self-verifying deploy (12 checks).

**ShopStream — AWS Big Data Pipeline** | May – Jul. 2026
*PySpark · AWS Lambda · EMR · Glue · RDS PostgreSQL · Flask · Zappa*

- Built an end-to-end pipeline for 2.5M e-commerce events: partitioned S3 ingestion, Lambda validation (CloudWatch metrics and quarantine), EMR/PySpark analytics covering six KPIs and anomaly detection, Glue ETL, and loading into an RDS PostgreSQL warehouse.
- Reached 86% test coverage with pytest, automated CI/CD with GitHub Actions, and hardened RDS security groups for controlled access.

**AWS Big Data Platform (Chinook / NYC Taxi)** | Mar. – May 2026 | Team of 3
*EC2 · RDS · Athena · S3 · Glue · Terraform · FastAPI · React*

- Processed the NYC Yellow Taxi dataset (94M rows, 12 months) on EMR with PySpark, resolving schema inconsistencies across Parquet files via explicit StructType and unionByName normalization.
- Deployed a full-stack application (React/Vite and FastAPI on two EC2 instances, PostgreSQL on RDS) and a Hive-partitioned external table in Athena over S3.

**Ayuda Terremoto Colombia — Distributed Humanitarian Coordination** | Aug. 2026 – Present
*FastAPI · PostgreSQL · React · Vite · IndexedDB · Render*

- Developed an offline-first humanitarian coordination platform following the magnitude 7.4 earthquake in Chocó: FastAPI/PostgreSQL backend as the central node and a React/IndexedDB local node with a per-action retry sync queue.
- Implemented fuzzy-matching report deduplication, automatic activation via USGS polling, and HXL-format export; covered the backend with 80+ pytest tests.

**NEXUS — Personal AI Assistant** | 2025 – Present (v2 since Oct. 2026)
*Python · MCP · SQLite FTS5 · systemd · Whisper · Chatterbox · Tailscale · GCP*

- Rebuilt the assistant as a ~8,800-line layer on Phantom (1,000+ tests, no root access) after a ~47,000-line v1 that was never reliable enough: a daily briefing, an agenda that turns emails into Google Calendar events, push-to-talk voice, and memory shared by Claude Code, OpenCode, and Hermes over MCP.
- Built a copilot that fixes test-breaking commits in an isolated worktree and merges only after Claude's review, a natural-language gym log over Telegram with deterministic load progression, and a watchdog on a free-tier cloud VM that alerts on service outages.

Additional projects: Victor (Jul. – Aug. 2026), a voice race engineer for Linux sim racing (C shared-memory reader, failover across 4 LLM providers, 143 tests); a Sentiment140 sentiment-analysis lab tracked in MLflow (team of 5, champion model with 0.824 macro-F1 served by FastAPI on EC2, Sep. 2026); Dino Island, a generative NLP pipeline (team of 5: character-level GRU, gemma4 via Ollama on SageMaker, aMUSEd images, Lambda app, Oct. 2026); an NLP API with spaCy and FastAPI on AWS Lambda and EC2 (93/100 in a live evaluation); a custom programming language built with ANTLR4; RL and CNN labs.

## TECHNICAL SKILLS

**Languages:** Python (intermediate-advanced), SQL, C, Bash; TypeScript, Rust, and JavaScript (basic)

**Machine Learning and AI:** scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, spaCy, MLflow, Whisper, Ollama

**Big Data and Cloud:** AWS (EC2, S3, RDS, Athena, Glue, EMR, Lambda, SageMaker), PySpark, Terraform, Docker (basic)

**Data and ETL:** Pandas, NumPy, Matplotlib, Power BI, Jupyter Notebook, openpyxl

**Backend and Web:** FastAPI, Flask, Express, React, SQLAlchemy, Prisma

**Tooling:** Git, GitHub Actions, pytest, ChromaDB, MCP, LLM APIs (Claude, Groq, Cerebras, NVIDIA NIM, Gemini)

## ACHIEVEMENTS

**1st Place — Wisibilízalas International Competition, Universitat Pompeu Fabra** | 2019

- Led the winning web application as technical product manager and frontend/UX lead, representing Ecuador in the Senior category at age 17 against teams from Latin America and Europe.

**1st Place — USABOT Robotics Challenge** | November 2025

- Directed the CAD design and sensor architecture of a smart automated irrigation system and delivered the commercial pitch to the judges, winning against university teams.

## CERTIFICATIONS

AWS Academy Graduate — Data Engineering | Amazon Web Services Academy | May 2026 (40 hours)

Database Programming with SQL | Oracle Academy | December 2024

Database Design | Oracle Academy | August 2024

## LANGUAGES

Spanish: native. English: certified B2 overall (iTEP Academic-Plus, November 2025; Listening C1, Reading and Speaking B2).